

Polynomials

6

SYLLABUS

1.11

2.12

HL ONLY

Nature rarely changes in sudden jumps. Temperature, water level and the position of a moving object all change smoothly, and polynomials are the simplest functions that change in the same way. The degree matches the situation: distance travelled at a constant speed is linear, a thrown ball follows a parabola, and the volume of a sphere grows with the cube of its radius. Over a short enough interval almost any smooth curve can be matched closely by a polynomial, so measurements are often fitted with one when no exact formula is known.

See Ch. 16.

That smoothness comes from the way a polynomial is built. It uses one variable and a few constants, combined by addition and multiplication only. There is no division by the variable, no square root and no trigonometry. A polynomial is therefore defined for every real number, so its graph has no gaps, no asymptotes and no corners.

Describing a polynomial is easy. Solving one is not. An exact formula for the roots exists for degree 2, degree 3 and degree 4, and Niels Abel proved in 1824 that for degree 5 and above no such formula exists. By the end of this chapter you will be able to read a polynomial's graph from its factorised form, divide one polynomial by another, test whether a value is a root and factorise once you have found one, and give the sum and the product of the roots from the coefficients alone.

One idea underlies all of it: a polynomial's coefficients and its roots describe the same object, and nearly every technique here moves between the two. The chapter closes with **partial fractions**, which splits a fraction by factorising its denominator and giving each factor a numerator of its own. That is why it belongs here rather than with the rational functions of §5.4, and it is what makes a rational function possible to integrate later. See Ch. 15.

6.1 Polynomial Functions and their Graphs

Much of the graph of a polynomial can be described before a single point is plotted, because the degree and the coefficients already carry that information. Reading those features from the equation is faster than building a table of values, and they are what a sketch has to show.

DEF A **polynomial function** of **degree** n has the form

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0,$$

where the coefficients are constants and n is a non-negative integer.

6.1.1 Degree, Coefficients and End Behaviour

The graph features in this subsection are read from two things: the degree, and the sign of the lead coefficient.

Polynomials are named by their degree. Degree 1 is **linear**, degree 2 is **quadratic**, degree 3 is **cubic**, degree 4 is **quartic**, and degree 5 is **quintic**. The degree also limits the shape: it limits how many times the curve can meet the x -axis, and how often it can change direction.

The constants $a_n, a_{n-1}, \dots, a_1, a_0$ are the **coefficients** of the powers of x . The coefficient of the highest power is a_n , called the **lead coefficient** (also written leading coefficient), and the term $a_n x^n$ that contains it is the **leading order term**.

For a factorised polynomial neither has to be multiplied out. The degree is the total number of linear factors, and the lead coefficient is the product of the leading coefficients of those factors.

A coefficient may take any value, with one restriction: the lead coefficient cannot be zero. If a_n were zero, the term $a_n x^n$ would be zero, and what remained would be a polynomial of degree $n - 1$ instead.

The sign of the lead coefficient determines which way the ends of the curve point. A polynomial with $a_n > 0$ is sometimes called a **positive polynomial**, and one with $a_n < 0$ a **negative polynomial**. These names describe the lead coefficient only. They do not mean that the function is positive or negative everywhere: $y = x^2 - 4$ has a positive lead coefficient, yet y is negative for $-2 < x < 2$.

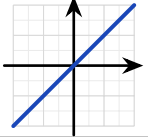
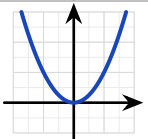
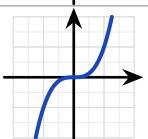
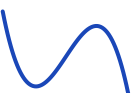
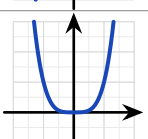
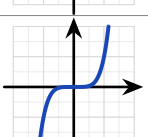
Three features are visible before you plot a single point.

The **end behaviour**, meaning the direction of the curve as $x \rightarrow \pm\infty$, is determined entirely by the leading term. If n is odd, the two ends point in opposite directions. If n is even, they point the same way, and the sign of a_n determines whether that direction is upward or downward.

The **roots** are the places where the graph meets the x -axis, and their number can never exceed the degree. A cubic meets the axis at most three times, a quartic at most four.

A **turning point** is a place where the curve stops rising and begins to fall, or stops falling and begins to rise. A polynomial of degree n has at most $n - 1$ turning points. The possible numbers then decrease in steps of two: $n - 1$, $n - 3$, and so on. The number changes by two rather than by one because a maximum and a minimum are always lost together, never one alone.

The table sets out all three features for the degrees you meet most often.

n	$y = x^n$	Shape when $a_n > 0$	Shape when $a_n < 0$	Times it meets the x -axis	Turning points
1				1	0
2				0, 1 or 2	1
3				1, 2 or 3	0 or 2
4				0, 1, 2, 3 or 4	1 or 3
5				1, 2, 3, 4 or 5	0, 2 or 4

EXAMPLE 6.1

State the degree, the lead coefficient and the two end behaviours of (a) $p(x) = 5 - 4x + 6x^3 - x^2$ and (b) $q(x) = (1 - 2x)(x + 3)^2(x - 4)$.

(a) The terms are not written in order of power, so find the highest power rather than reading the first term. It is $6x^3$, so the degree is 3 and the lead coefficient is 6. An odd degree points the two ends in opposite directions, and a positive lead coefficient sends the right-hand end upward:

$$p(x) \rightarrow +\infty \text{ as } x \rightarrow +\infty, \quad p(x) \rightarrow -\infty \text{ as } x \rightarrow -\infty.$$

(b) Nothing needs to be multiplied out. Count the linear factors: $(1 - 2x)$, two copies of $(x + 3)$,

and $(x-4)$, so the degree is 4. The lead coefficient is the product of the four leading coefficients,

$$(-2) \times 1 \times 1 \times 1 = -2.$$

An even degree points both ends the same way, and the negative lead coefficient sends them down: $q(x) \rightarrow -\infty$ as $x \rightarrow \pm\infty$.

Everything in (b) turns on the -2 , which comes from the factor written backwards as $(1-2x)$. Reading the coefficient of x in that bracket as 2 and losing the minus sign gives a lead coefficient of $+2$, which reverses both ends.

6.1.2 Roots and Multiplicity

The table gives the shape in outline. What fixes it exactly is what the curve does at each root, and that is determined by whether a factor is repeated.

The **multiplicity** of a root is the number of times its factor appears in the factorised polynomial. In $(x-1)(x-2)^2$ the root $x=1$ has multiplicity one, and the root $x=2$ has multiplicity two. A root of multiplicity one is called a **simple root**, and one of multiplicity two a **double root**.

KEY At a root of **multiplicity one** the curve crosses the axis straight. At a root of **even** multiplicity it touches the axis and turns back. At a root of **odd** multiplicity greater than one it crosses, but flattens against the axis as it does so.

Whether the multiplicity is odd or even determines crossing or touching. Its size determines how flat the curve is where it meets the axis.

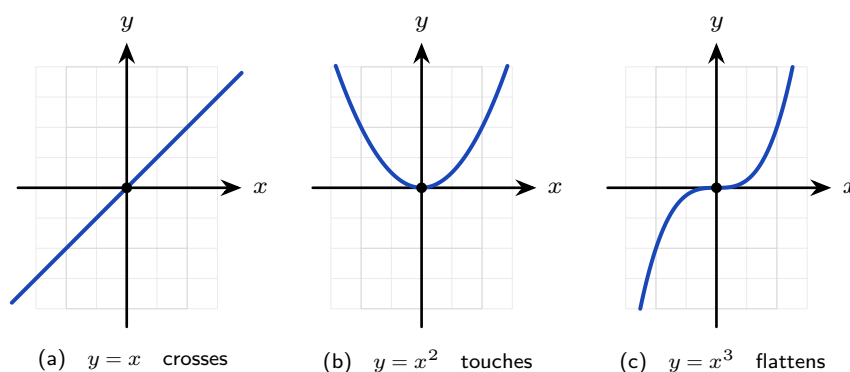


Figure 6.1 The three behaviours at a root, shown at $x=0$. In (a) the multiplicity is one and the curve crosses straight. In (b) it is two, an even number, so the curve touches and turns back. In (c) it is three, odd, so the curve crosses again, but flattens against the axis as it passes through.

Three readings then give the whole sketch. Take the behaviour at each x -intercept from the roots and their multiplicities, the y -intercept from the value at $x=0$, and the two ends from the leading term.

Practise both directions: reading the multiplicities from a sketch, and predicting the sketch

from the factors.

EXAMPLE 6.2

Sketch $y = (x + 1)(x - 2)^2$, showing the intercepts and the behaviour at each root.

Read the roots and their multiplicities from the factors. The factor $(x + 1)$ is single, so the graph crosses the axis at $x = -1$. The factor $(x - 2)^2$ is squared, so the graph touches the axis at $x = 2$ and turns back.

Find the y -intercept. Substitute $x = 0$: $y = (1)(-2)^2 = 4$, the point $(0, 4)$.

Fix the ends from the leading term. Multiplying out leads with x^3 : an odd degree with a positive coefficient, so the curve runs from $-\infty$ at the left to $+\infty$ at the right.

These three facts fix the shape.

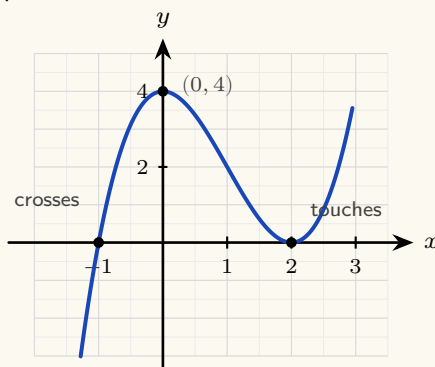


Figure 6.2 A sketch of $y = (x + 1)(x - 2)^2$. The curve crosses the axis at the simple root $x = -1$ and touches it at the double root $x = 2$. The y -intercept is $(0, 4)$, and the ends follow x^3 .

Every factorised polynomial is sketched the same way: the roots and their multiplicities give the behaviour at the x -axis, the value at $x = 0$ gives the y -intercept, and the leading term gives the two ends.

When the polynomial is written out in full rather than factorised, the y -intercept is even quicker to read. Every term except the constant contains x , so substituting $x = 0$ makes all of them zero. The constant term a_0 is what remains.

KEY The graph of $y = a_n x^n + \dots + a_1 x + a_0$ crosses the y -axis at $(0, a_0)$. The constant term is the y -intercept.

For $y = 2x^3 - 3x + 5$, the constant term is 5, so the graph crosses the y -axis at $(0, 5)$.

6.1.3 From the Graph Back to the Equation

The reverse problem is also examined. Given a sketch, you are asked for the polynomial that produced it. The intercepts give you almost everything you need.

Each x -intercept contributes a factor, and the behaviour there gives its multiplicity: a crossing gives a single factor, a touch gives a squared factor. That fixes the polynomial up to one unknown number, the leading coefficient, and any known point that is not an x -intercept determines it. Check that the factors account for the whole degree of the curve; if they do not,

an intercept has been missed or a multiplicity is higher than it looks.

KEY Finding a polynomial from its graph. Write one factor for each x -intercept, with the multiplicity the graph shows at that point. Then substitute the coordinates of any known point that is not an x -intercept to find the leading coefficient a_n .

EXAMPLE 6.3

The graph below shows a cubic. It touches the x -axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$. Find its equation.

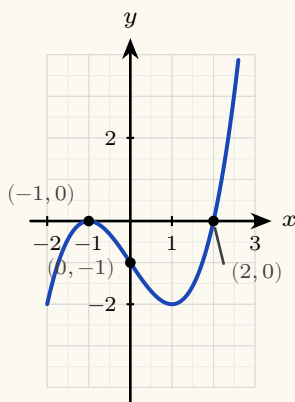


Figure 6.3 The curve touches the x -axis at $x = -1$, crosses it at $x = 2$, and passes through $(0, -1)$.

These three facts determine the equation.

Write the factors from the x -intercepts. The curve touches at $x = -1$, so that root is repeated and contributes $(x+1)^2$. It crosses at $x = 2$, so that root is single and contributes $(x-2)$. Those factors already account for degree three, so

$$y = a(x+1)^2(x-2)$$

for some constant a .

Find a from the remaining point. Substitute $x = 0$ and $y = -1$:

$$-1 = a(1)^2(-2) = -2a,$$

so $a = \frac{1}{2}$ and

$$y = \frac{1}{2}(x+1)^2(x-2).$$

Check the answer against the graph. The leading term is $\frac{1}{2}x^3$, which is positive, so the curve rises to the right, and the sketch agrees.

The intercepts fix the factors; one further point fixes the leading coefficient.

YOUR TURN 6.1 Polynomial Functions and their Graphs

1. State the degree and the lead coefficient of each polynomial.

(a) $f(x) = 3x^4 - 2x + 7$

(b) $f(x) = 5 - x^3$

(c) $f(x) = (x - 1)(x + 2)(x - 3)$

(d) $f(x) = -2x^5 + x^2$

2. Describe the end behaviour of each graph as $x \rightarrow +\infty$ and as $x \rightarrow -\infty$.

(a) $y = x^3$

(b) $y = -x^4$

(c) $y = 2x^5 - x$

(d) $y = 3 - x^2$

3. Answer each question about degree and shape.

(a) State the maximum number of real roots of a degree-5 polynomial.

(b) State the maximum number of turning points of a quartic.

(c) List every possible number of turning points a cubic can have.

(d) Explain why a cubic must have at least one real root.

4. State the roots of each polynomial, and say at each one whether the graph crosses or touches the x -axis.

(a) $y = (x - 2)(x + 3)(x - 5)$

(b) $y = (x - 1)^2(x + 4)$

(c) $y = x(x - 3)^2$

(d) $y = (x + 1)^2(x - 2)^2$

5. Consider $y = (x + 1)(x - 3)^2$.

(a) State the roots and their multiplicities.

(b) Find the y -intercept.

(c) State the end behaviour.

(d) Sketch the curve.

6. A cubic touches the x -axis at $x = 2$, crosses it at $x = -1$, and passes through $(0, 4)$.

(a) Write down the factors, with the correct multiplicities.

(b) Find the lead coefficient.

(c) Write the equation of the cubic.

6.2 Polynomial Division

You can already add, subtract and multiply polynomials. Addition and subtraction collect like terms, and multiplication expands every product before collecting. Division is the operation that still needs a method, and it is the one §6.3 and §6.4 are built on.

Factorising a polynomial means writing it as a product. Once you know one factor, the other is found by dividing. So a method for dividing one polynomial by another is needed before a higher-degree polynomial can be factorised from a known factor.

6.2.1 The Division Algorithm

Before dividing anything it helps to know what the answer should look like, and whole-number division already shows that.

Polynomial division follows the same pattern. When you divide 47 by 5 you write

$$47 = 5 \times 9 + 2,$$

which records a divisor, a quotient, and a remainder smaller than the divisor. Polynomial division works the same way, with “smaller” measured by degree instead of size.

KEY **The division algorithm.** For polynomials $f(x)$ and $d(x)$, with $d(x)$ not the zero polynomial, there are unique polynomials $q(x)$ and $r(x)$ such that

$$f(x) = d(x)q(x) + r(x),$$

where the degree of r is less than the degree of d . *Dividend equals divisor times quotient, plus remainder.*

Fix the four names now, because the methods that follow use them. The polynomial being divided, f , is the **dividend**. The polynomial you divide by, d , is the **divisor**. The result q is the **quotient**, and the part that remains, r , is the **remainder**. When $r = 0$ the division is exact, and d is a factor of f .

6.2.2 Long Division

Long division is the method that works for a divisor of any degree. It is three steps repeated until the remainder has lower degree than the divisor.

KEY Divide, multiply, subtract.

1. **Divide** the leading term of the dividend by the leading term of the divisor. The result is the next term of the quotient.
2. **Multiply** the whole divisor by that term.
3. **Subtract** the product from the dividend, and repeat on what is left.

Set the work out in columns, one for each power of x : the divisor to the left of the bracket, the dividend inside it, the quotient written on the top line, and the remainder on the bottom line. Write the result as $f(x) = d(x)q(x) + r(x)$ when you finish, because multiplying that right-hand side out checks the whole division.

EXAMPLE 6.4

Divide $2x^3 - 3x^2 + x - 5$ by $(x - 2)$.

Set the work out in columns, one column for each power of x .

$$\begin{array}{r}
 \text{ quotient} \\
 x-2 \text{ dividend} \\
 \underline{2x^3 - 4x^2} \\
 x^2 + x \\
 \underline{ x^2 - 2x} \\
 3x - 5 \\
 \underline{ 3x - 6} \\
 1 \text{ remainder}
 \end{array}$$

The divisor $x-2$ is written to the left of the bracket, and the dividend $2x^3-3x^2+x-5$ inside it. The quotient is written on the top line, one term at each stage, and the remainder appears on the bottom line.

Each stage repeats the same three steps: divide, multiply, subtract.

Stage 1. Divide the leading term of the dividend by the leading term of the divisor: $2x^3 \div x = 2x^2$. Write $2x^2$ in the quotient. Multiply the divisor by it: $2x^2(x-2) = 2x^3-4x^2$. Subtract that from the dividend, which leaves $x^2 + x$.

Stage 2. Repeat on $x^2 + x$. Divide: $x^2 \div x = x$. Multiply: $x(x-2) = x^2 - 2x$. Subtract, which leaves $3x - 5$.

Stage 3. Repeat on $3x - 5$. Divide: $3x \div x = 3$. Multiply: $3(x-2) = 3x - 6$. Subtract, which leaves 1.

The degree of 1 is lower than the degree of $x-2$, so the division stops.

The quotient is $2x^2 + x + 3$ and the remainder is 1:

$$2x^3 - 3x^2 + x - 5 = (x - 2)(2x^2 + x + 3) + 1.$$

Always write the answer in this form. Multiplying out the right side is a complete check of the di-

vision.

CAUTION Write the dividend in descending powers and insert a zero term for every missing power before dividing. Divide $x^3 - 13x + 12$ as $x^3 + 0x^2 - 13x + 12$. Without that zero term the columns shift, and the subtractions that follow go wrong.

EXAMPLE 6.5

Divide $x^3 - 13x + 12$ by $(x - 1)$.

There is no x^2 term, so write it as $x^3 + 0x^2 - 13x + 12$ and keep a column for it.

$$\begin{array}{r}
 \overline{x^2 + x - 12} \\
 x-1 \overline{x^3 + 0x^2 - 13x + 12} \\
 \underline{x^3 - x^2} \\
 - 13x \\
 \underline{x^2 - x} \\
 - 12x + 12 \\
 \underline{-12x + 12} \\
 0
 \end{array}$$

The remainder is 0, so the division is exact and $(x - 1)$ is a factor. The quotient $x^2 + x - 12$ factorises further:

$$x^3 - 13x + 12 = (x - 1)(x^2 + x - 12) = (x - 1)(x + 4)(x - 3).$$

A remainder of zero shows that the divisor is a factor. The next two sections are built on that fact.

A divisor of higher degree changes none of the three steps. What it changes is where the division stops: the remainder must have degree lower than the divisor, so dividing by a quadratic can leave a remainder with an x in it. Set the columns out in the same way, and keep a column for every power of x from the highest down to the constant, whether or not the dividend has a term there.

EXAMPLE 6.6

Divide $x^4 + 2x^2 - 3x + 1$ by $(x^2 + x - 1)$.

There is no x^3 term, so write the dividend as $x^4 + 0x^3 + 2x^2 - 3x + 1$ and give that power a col-

umn of its own.

$$\begin{array}{r}
 \overline{x^2 - x + 4} \\
 x^2 + x - 1 \overline{) x^4 + 0x^3 + 2x^2 - 3x + 1} \\
 \underline{x^4 + x^3 - x^2} \\
 -x^3 + 3x^2 - 3x \\
 \underline{-x^3 - x^2 + x} \\
 4x^2 - 4x + 1 \\
 \underline{4x^2 + 4x - 4} \\
 -8x + 5
 \end{array}$$

Stage 1. Divide the leading terms: $x^4 \div x^2 = x^2$. Multiply the divisor by it: $x^2(x^2 + x - 1) = x^4 + x^3 - x^2$. Subtract, which leaves $-x^3 + 3x^2 - 3x$.

Stage 2. Repeat. Divide: $-x^3 \div x^2 = -x$. Multiply: $-x(x^2 + x - 1) = -x^3 - x^2 + x$. Subtract, which leaves $4x^2 - 4x + 1$.

Stage 3. Repeat. Divide: $4x^2 \div x^2 = 4$. Multiply: $4(x^2 + x - 1) = 4x^2 + 4x - 4$. Subtract, which leaves $-8x + 5$.

The remainder $-8x + 5$ has degree one, the divisor has degree two, so the division stops:

$$x^4 + 2x^2 - 3x + 1 = (x^2 + x - 1)(x^2 - x + 4) - 8x + 5.$$

Multiplying the right side out returns the dividend, which checks every stage at once. A remainder with an x in it is the normal outcome against a quadratic divisor, not a sign that a stage has gone wrong.

YOUR TURN 6.2 Polynomial Division

7. Simplify each expression.

(a) $(x^3 + 2x - 5) + (3x^3 - x^2 + 1)$

(b) $(x^4 + 3x^2 - 1) - (2x^3 - x^2 + 2)$

(c) $(x^2 + 2x - 1)(x^2 - x + 3)$

(d) $(x - 2)(x^2 + 2x + 4)$

8. Divide, and state the quotient and the remainder.

(a) $x^2 + 5x + 6$ by $(x + 2)$

(b) $x^3 - 3x^2 + 2x$ by $(x - 1)$

(c) $x^3 + 2x^2 - x + 1$ by $(x - 1)$

(d) $x^3 - 8$ by $(x - 2)$

9. Divide, and state the quotient and the remainder. Insert a zero term for each missing power first.

(a) $2x^3 - x^2 - 3$ by $(x + 1)$

(b) $x^3 + 3x^2 - 4$ by $(x + 2)$

(c) $2x^4 - 3x^3 + x - 6$ by $(x - 2)$

10. Divide by the quadratic divisor.

(a) $x^4 - 1$ by $(x^2 - 1)$

(b) $x^4 - 1$ by $(x^2 + 1)$

6.3 The Remainder and Factor Theorems

6.3.1 The Remainder Theorem

Here are two operations that appear unrelated. *Evaluating* f at 2 gives the height of the graph above one point. *Dividing* f by $(x - 2)$ is the long calculation of the previous section. The remainder theorem states that the two produce the same number, and it is quickly proved.

Take the division algorithm with a linear divisor $(x - a)$:

$$f(x) = (x - a)q(x) + r,$$

where the remainder r must be a *constant*, because its degree is less than the degree of $x - a$, and less than one means zero.

Now substitute $x = a$ into that identity. The factor $(x - a)$ becomes zero, so the whole first term is zero, leaving $f(a) = r$. The height of the graph at a is the remainder, because everything else in the polynomial is divisible by $(x - a)$ and so contributes nothing at that point.

KEY **Remainder theorem.** When a polynomial $f(x)$ is divided by $(x - a)$, the remainder is $f(a)$. *One substitution replaces the division.*

The theorem holds for any linear divisor, not only for one written as $(x - a)$. Divide by $(bx - c)$ and the remainder is f evaluated at the **zero of the divisor**, meaning the value of x that makes the divisor equal to zero. For $(2x - 1)$ that value is $\frac{1}{2}$, never 1 or 2.

Read the theorem in either direction. Given the coefficients it returns the remainder; given two remainders it returns two linear equations in the unknown coefficients.

EXAMPLE 6.7

Find the remainder when $f(x) = x^3 - 2x + 5$ is divided by $(x - 2)$.

By the remainder theorem, the remainder is

$$f(2) = 2^3 - 2(2) + 5 = 8 - 4 + 5 = 9.$$

No division needed; the answer is a single evaluation.

EXAMPLE 6.8

Find the remainder when $f(x) = 8x^3 + 4x^2 - 6x + 5$ is divided by $(2x - 1)$.

The divisor need not have the form $(x - a)$. The theorem works for any linear divisor, provided you evaluate at the **zero of the divisor**, meaning the value of x that makes the divisor equal to zero.

Here $2x - 1 = 0$ when $x = \frac{1}{2}$, so the remainder is

$$f\left(\frac{1}{2}\right) = 8 \cdot \frac{1}{8} + 4 \cdot \frac{1}{4} - 3 + 5 = 1 + 1 - 3 + 5 = 4.$$

Evaluate at the zero of the divisor, not at one of its coefficients. Dividing by $(2x - 1)$ means substituting $\frac{1}{2}$, never 1 or 2.

EXAMPLE 6.9

The polynomial $f(x) = x^3 + ax + b$ leaves remainder 7 when divided by $(x - 2)$ and remainder -5 when divided by $(x + 1)$. Find a and b .

Each remainder condition is one evaluation, so two conditions give two equations:

$$f(2) = 8 + 2a + b = 7 \implies 2a + b = -1,$$

$$f(-1) = -1 - a + b = -5 \implies -a + b = -4.$$

Subtracting, $3a = 3$, so $a = 1$ and $b = -3$. The remainder theorem has turned two divisions into a pair of linear equations; questions of this kind, working from the remainders back to the coefficients, are common.

The theorem is stated for a linear divisor, but the idea behind it reaches further. Divide by a quadratic and the remainder has degree at most one, so it is $rx + s$: two unknowns instead of one. The division algorithm still records the whole calculation as an identity, and each zero of the divisor makes the divisor term vanish, leaving one equation in r and s . Two distinct zeros give two equations, which is exactly what two unknowns need.

EXAMPLE 6.10

Find the remainder when $f(x) = x^3 + 2x^2 - 5x + 3$ is divided by $(x - 1)(x - 2)$.

The divisor has degree two, so the remainder has degree at most one. Write it as $rx + s$ and record the division as an identity:

$$f(x) = (x - 1)(x - 2)q(x) + rx + s.$$

This holds for every value of x , so choose the two values that make the product term disappear. Substituting $x = 1$,

$$f(1) = 1 + 2 - 5 + 3 = 1 \implies r + s = 1,$$

and substituting $x = 2$,

$$f(2) = 8 + 8 - 10 + 3 = 9 \implies 2r + s = 9.$$

Subtracting the first equation from the second gives $r = 8$, and then $s = -7$. The remainder is

$$8x - 7.$$

Long division confirms it: the quotient is $x + 5$, and $(x - 1)(x - 2)(x + 5) + 8x - 7$ returns $f(x)$. The method depends on the divisor having two *distinct* zeros; against $(x - 1)^2$ there is only one substitution available, and the second equation has to come from comparing coefficients instead.

6.3.2 The Factor Theorem

More follows when the remainder is zero. If dividing by $(x - a)$ leaves no remainder, then $(x - a)$ divides $f(x)$ exactly: it is a factor. That is what turns a test for a root into a method for factorising.

KEY **Factor theorem.** $(x - a)$ is a factor of $f(x)$ if and only if $f(a) = 0$.

To factorise an unfamiliar cubic, search for a root by testing small values. The factors of the constant term are the natural candidates, and the reason is short. If a is an integer root of $a_n x^n + \dots + a_1 x + a_0$ with integer coefficients, then substituting and moving a_0 across gives

$$a(a_n a^{n-1} + a_{n-1} a^{n-2} + \dots + a_1) = -a_0,$$

so a divides a_0 . When the leading coefficient is not 1, the candidates are the fractions $\frac{p}{q}$ with p a factor of a_0 and q a factor of a_n . Once one root a is found, $(x - a)$ is a factor, and dividing it out leaves a quadratic you can finish by hand.

That division need not be long division. To **compare coefficients**, write the quotient with unknown coefficients, one degree lower than the dividend, multiply out, and equate the coefficients of matching powers. Each unknown comes from one comparison, and any coefficient left over is a check.

EXAMPLE 6.11

Show that $(x - 1)$ is a factor of $f(x) = x^3 - 7x^2 + 14x - 8$, and factorise f completely.

Test $x = 1$:

$$f(1) = 1 - 7 + 14 - 8 = 0,$$

so by the factor theorem $(x - 1)$ is a factor.

Now divide it out. Instead of long division, *compare coefficients*. The quotient must be a quadratic, so write

$$x^3 - 7x^2 + 14x - 8 = (x - 1)(x^2 + bx + c),$$

then compare the coefficients of matching powers on the two sides. Compare the constant terms: $-c = -8$, so $c = 8$. Compare the coefficients of x^2 : $b - 1 = -7$, so $b = -6$. The coefficient of x has not been used, so use it as a check: on the right it is $c - b = 8 + 6 = 14$, which matches the left. ✓ Hence

$$f(x) = (x - 1)(x^2 - 6x + 8) = (x - 1)(x - 2)(x - 4).$$

The constant term -8 gives the candidates $\pm 1, \pm 2, \pm 4, \pm 8$. Testing $x = 1$ succeeded on the first attempt, and comparing coefficients carried out the division with a check included.

TIP Comparing coefficients is usually faster than long division in an examination. Each unknown comes from one comparison, and any coefficient you did not use provides a check. Long division is equally acceptable if you prefer it.

YOUR TURN 6.3 The Remainder and Factor Theorems

11. Find the remainder in each case.

- | | |
|--|---|
| (a) $x^3 + 2x - 1$ divided by $(x - 1)$ | (b) $x^3 - 4x^2 + x + 6$ divided by $(x - 2)$ |
| (c) $2x^3 - 3x + 1$ divided by $(x + 1)$ | (d) $x^3 + x^2 + x + 1$ divided by $(x + 2)$ |

12. Find the remainder in each case. The divisor is not of the form $(x - a)$.

- | | |
|--|--|
| (a) $4x^3 - 4x^2 + 5x - 1$ divided by $(2x - 1)$ | (b) $2x^3 + x^2 - 2$ divided by $(2x + 1)$ |
|--|--|

13. Interpret each statement.

- | | |
|--|--|
| (a) The remainder when $f(x)$ is divided by $(x - 3)$ is 0. What does this tell you? | (b) The remainder when $f(x)$ is divided by $(x - 2)$ is 7. State $f(2)$. |
| (c) $f(-1) = 0$. Write down a factor of $f(x)$. | (d) $x^4 - 1$ is divided by $(x - 1)$. Find the remainder without dividing. |

14. The polynomial $f(x) = x^3 + ax + b$ leaves remainder 2 when divided by $(x - 1)$ and remainder -13 when divided by $(x + 2)$.

- (a) Write down two equations in a and b .
 (b) Solve them to find a and b .

15. Decide whether the given linear expression is a factor. Show your working.

- | | |
|--|---|
| (a) Is $(x - 2)$ a factor of $x^3 - 3x^2 + 4$? | (b) Is $(x + 1)$ a factor of $x^3 + 2x^2 - x - 2$? |
| (c) Is $(x - 1)$ a factor of $x^3 + x^2 + x + 1$? | (d) Is $(x + 2)$ a factor of $x^3 + 8$? |

16. Factorise each polynomial completely.

(a) $x^3 - x$

(b) $x^3 - 4x$

(c) $x^3 + 2x^2 - x - 2$

(d) $2x^3 + x^2 - 5x + 2$

17. Answer each part.

- (a) One factor of $x^2 - 5x + 6$ is $(x - 2)$. State the other.
 (b) Show that $x^3 - 3x^2 + 4$ has a repeated factor.

6.4 Solving Polynomial Equations

You already know how to solve linear and quadratic equations. A linear equation is rearranged directly. A quadratic is solved by factorising, by completing the square, or by the quadratic formula.

For a cubic or higher there is no formula of that kind simple enough to use by hand, and from degree five upward none exists at all. The factor theorem takes its place.

KEY To solve a polynomial equation of degree 3 or more:

1. Test the divisors of the constant term until one value a gives $f(a) = 0$.
2. Divide $f(x)$ by $(x - a)$, either by long division or by comparing coefficients.
3. Repeat on the quotient until a quadratic remains, then solve that in the usual way.

Every real root is an x -intercept of the graph, so the algebra and the graph always agree.

EXAMPLE 6.12

Solve $x^3 - 3x^2 - x + 3 = 0$.

Test the divisors of 3. At $x = 1$: $1 - 3 - 1 + 3 = 0$, so $(x - 1)$ is a factor. Dividing,

$$x^3 - 3x^2 - x + 3 = (x - 1)(x^2 - 2x - 3) = (x - 1)(x - 3)(x + 1).$$

The roots are $x = 1, 3, -1$.

CAUTION A cubic with real coefficients always has at least one real root, because one end goes to $+\infty$ and the other to $-\infty$, so the curve must cross the axis. If no rational root can be found by testing, one root is irrational, and in this course the equation is then solved with a calculator.

YOUR TURN 6.4 Solving Polynomial Equations

18. Solve each equation. Each is already factorised or factorises at sight.

(a) $(x-1)(x+2)(x-4) = 0$

(b) $(x-3)^2(x+1) = 0$

(c) $x^3 = 4x$

(d) $x^3 + x^2 - 2x = 0$

19. Solve each equation, using the factor theorem to find the first root.

(a) $x^3 - 6x^2 + 11x - 6 = 0$

(b) $x^3 - 2x^2 - x + 2 = 0$

(c) $2x^3 + x^2 - 5x + 2 = 0$

20. Solve each equation.

(a) $x^4 - 5x^2 + 4 = 0$ (Hint: substitute $u = x^2$)

(b) $x^3 - 1 = 0$, giving only the real root, and explain why there is exactly one.

6.5 Sum and Product of Roots

The coefficients of a polynomial determine the sum and the product of its roots. You can therefore find those two values *without solving the equation at all*. This matters whenever the roots are irrational, or when the equation is too hard to solve, because the sum and the product remain easy to obtain.

6.5.1 The Quadratic Case

The pattern is easiest to see where there are only two roots to follow.

If $ax^2 + bx + c = 0$ has roots α and β , then the polynomial is $a(x - \alpha)(x - \beta)$, and the expansion gives the coefficients:

$$a(x - \alpha)(x - \beta) = a(x^2 - (\alpha + \beta)x + \alpha\beta) = ax^2 - a(\alpha + \beta)x + a\alpha\beta.$$

Compare with $ax^2 + bx + c$. The middle coefficient is $b = -a(\alpha + \beta)$ and the constant is $c = a\alpha\beta$. Solving each for the sum and the product gives

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

The minus sign on the sum is not a convention to memorise. It comes from the brackets, one $-\alpha$ and one $-\beta$.

6.5.2 Every Degree at Once

Nothing in that expansion depended on there being two brackets, so the same argument settles every degree, using the counting argument of Ch. 2. Expanding

$$a_n(x - r_1)(x - r_2) \cdots (x - r_n)$$

means choosing, from each bracket, either the x or the root, exactly as in the binomial theorem. To build the x^{n-1} term you take x from all brackets but one, so each root appears exactly once, each carrying a single minus sign: the coefficient equals $-(r_1 + r_2 + \cdots + r_n)$, the *sum*. To build the constant term you take the root from *every* bracket: all n roots multiplied together, carrying n minus signs, which is where the $(-1)^n$ comes from. So the outermost coefficients of a polynomial give the sum and the product of its roots.

KEY · IB For the equation $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0$,

$$\text{sum of roots} = -\frac{a_{n-1}}{a_n}, \quad \text{product of roots} = (-1)^n \frac{a_0}{a_n}.$$

The second coefficient is the sum of the roots with its sign changed; the constant term is their product.

These are the outermost of a whole family of relations (Vieta's formulas), but the sum and product are the two you will use most, for checking answers, for building equations with required roots, and for finding an unknown coefficient.

Reading the relations backwards builds an equation from its roots. A quadratic whose roots have sum S and product P is

$$x^2 - Sx + P = 0,$$

so an equation whose roots are a transformation of α and β is found by transforming the sum and the product. The roots themselves are never needed.

One more relation fills the gap between them. Building the x^{n-2} term means taking the root from exactly two brackets and x from the rest, so that coefficient is the sum of the roots multiplied together in pairs. For a cubic $a_3 x^3 + a_2 x^2 + a_1 x + a_0 = 0$ with roots α, β, γ , the three relations together are

$$\alpha + \beta + \gamma = -\frac{a_2}{a_3}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{a_1}{a_3}, \quad \alpha\beta\gamma = -\frac{a_0}{a_3}.$$

The syllabus requires only the first and the last. The middle relation is a short extension, and it is the one to use whenever a question involves all three roots at once.

EXAMPLE 6.13

The equation $2x^2 - 5x + 3 = 0$ has roots α and β . Find $\alpha + \beta$ and $\alpha\beta$ without solving, then verify.

Here $a = 2$, $b = -5$, $c = 3$, so

$$\alpha + \beta = -\frac{-5}{2} = \frac{5}{2}, \quad \alpha\beta = \frac{3}{2}.$$

Solving confirms it: the roots are 1 and $\frac{3}{2}$, whose sum is $\frac{5}{2}$ and product is $\frac{3}{2}$. The sum and product were available from the coefficients before the roots were found.

EXAMPLE 6.14

The cubic $x^3 - 2x^2 - 9x + 18 = 0$ has roots α, β, γ . Write down $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$.

With $a_3 = 1$, $a_2 = -2$, $a_0 = 18$ and degree $n = 3$,

$$\alpha + \beta + \gamma = -\frac{-2}{1} = 2, \quad \alpha\beta\gamma = (-1)^3 \frac{18}{1} = -18.$$

(The roots happen to be $-3, 2, 3$: sum 2, product -18 , as promised.)

6.5.3 Symmetric Combinations of the Roots

The sum and the product are more useful than they appear, because many other expressions in the roots can be rebuilt from them. That is what makes the method valuable when the roots are irrational: the examples below answer precise questions about the roots without ever finding one.

A **symmetric combination** of the roots is an expression whose value does not change when the roots are exchanged. Both $\alpha^2 + \beta^2$ and $\frac{1}{\alpha} + \frac{1}{\beta}$ are symmetric; $\alpha - \beta$ is not. Every symmetric combination of two roots can be written in terms of $\alpha + \beta$ and $\alpha\beta$. Two rearrangements cover most cases, and a sum of reciprocals reduces by putting it over a common denominator, $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$.

KEY

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \quad (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta.$$

Neither is a new formula. Each is the expansion of $(\alpha \pm \beta)^2$ rearranged.

EXAMPLE 6.15

The equation $x^2 - 3x + 1 = 0$ has roots α and β . Without solving, find $\alpha^2 + \beta^2$ and $\frac{1}{\alpha} + \frac{1}{\beta}$.

The roots are irrational, but their sum and product are not: $\alpha + \beta = 3$ and $\alpha\beta = 1$. For the sum of squares, use the first rearrangement above:

$$(\alpha + \beta)^2 = \alpha^2 + 2\alpha\beta + \beta^2 \implies \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 9 - 2 = 7.$$

For the sum of reciprocals, combine over a common denominator:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{3}{1} = 3.$$

Neither answer required a root. Both were rebuilt from the sum and the product alone.

EXAMPLE 6.16

With α, β as above, find a quadratic equation with integer coefficients whose roots are 2α and 2β . A quadratic is determined by the sum and product of its roots: $x^2 - (\text{sum})x + (\text{product}) = 0$. The new sum and product follow from the old ones:

$$2\alpha + 2\beta = 2(\alpha + \beta) = 6, \quad (2\alpha)(2\beta) = 4\alpha\beta = 4.$$

So the required equation is

$$x^2 - 6x + 4 = 0.$$

No root was ever computed: transforming an equation's roots means transforming their sum and product, nothing more.

EXAMPLE 6.17

The roots of $x^2 + kx + 12 = 0$ differ by 1. Find the possible values of k .

Let the roots be α and β with $\alpha - \beta = 1$. The coefficients give $\alpha + \beta = -k$ and $\alpha\beta = 12$. Connect the difference to these totals by squaring it:

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta \implies 1 = k^2 - 48,$$

so $k^2 = 49$ and $k = \pm 7$. Check with $k = 7$: the equation $x^2 + 7x + 12 = 0$ has roots -3 and -4 , which indeed differ by 1. A condition on the roots has become an equation in the coefficient.

YOUR TURN 6.5 Sum and Product of Roots

21. State the sum and the product of the roots of each equation.

(a) $x^2 - 7x + 10 = 0$

(b) $2x^2 + 3x - 5 = 0$

(c) $x^2 + 4x + 4 = 0$

(d) $3x^2 - 2x - 8 = 0$

22. For each cubic or quartic, state the sum and the product of the roots.

(a) $x^3 - 6x^2 + 11x - 6 = 0$

(b) $2x^3 - 4x^2 + x - 3 = 0$

(c) $x^4 - 3x^3 + 2x - 5 = 0$

23. Build the required equation.

- (a) Write a quadratic with lead coefficient 1 whose roots sum to 5 and multiply to 6.
- (b) The roots of $x^2 + bx + 12 = 0$ are 3 and one other. Find the other root and b .
- (c) Find the value of k for which $x^2 - kx + 9 = 0$ has equal roots.
- (d) Write a cubic with lead coefficient 1 whose roots are -1 , 2 and 4.

24. The roots of $2x^2 - 7x + 6 = 0$ are α and β . Without solving the equation, find each value.

- (a) $\alpha + \beta$ and $\alpha\beta$
- (b) $\frac{1}{\alpha} + \frac{1}{\beta}$
- (c) $\alpha^2 + \beta^2$
- (d) $(\alpha - \beta)^2$

25. The roots of $x^2 + 5x + 3 = 0$ are α and β .

- (a) Write down $\alpha + \beta$ and $\alpha\beta$.
- (b) Find the sum and product of $\alpha + 2$ and $\beta + 2$.
- (c) Hence find a quadratic equation with integer coefficients whose roots are $\alpha + 2$ and $\beta + 2$.

26. The roots of $x^3 - 4x^2 + x + 6 = 0$ are α , β and γ .

- (a) Write down $\alpha + \beta + \gamma$ and $\alpha\beta\gamma$.
- (b) Write down $\alpha\beta + \beta\gamma + \gamma\alpha$.
- (c) Hence find $\alpha^2 + \beta^2 + \gamma^2$.

6.6 Partial Fractions

Adding two simple fractions produces one complicated fraction. **Partial fractions** reverse that process: they split a single fraction back into a sum of simpler ones, one for each factor of the denominator. The split form is easier to use, and it is what makes a rational function possible to integrate. See Ch. 15.

The syllabus limits the problem to a denominator of **at most two distinct linear factors**, with the numerator of lower degree than the denominator. That is the case treated below. The larger theory is sketched at the end of the section for reference.

The denominator is a product of two different linear factors, and each factor contributes one term with a constant numerator.

KEY For a proper fraction over two distinct linear factors,

$$\frac{px + q}{(x - \alpha)(x - \beta)} = \frac{A}{x - \alpha} + \frac{B}{x - \beta}.$$

To find A and B , multiply through by the denominator, then substitute the values of x that make one factor zero. Each substitution isolates one unknown.

EXAMPLE 6.18

Express $\frac{3x + 5}{(x - 1)(x + 2)}$ in partial fractions.

Write

$$\frac{3x + 5}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2},$$

and multiply through by $(x - 1)(x + 2)$:

$$3x + 5 = A(x + 2) + B(x - 1).$$

Setting $x = 1$ isolates A : $8 = 3A$, so $A = \frac{8}{3}$. Setting $x = -2$ isolates B : $-1 = -3B$, so $B = \frac{1}{3}$. Therefore

$$\frac{3x + 5}{(x - 1)(x + 2)} = \frac{8}{3(x - 1)} + \frac{1}{3(x + 2)}.$$

Choosing the two values that each make one factor zero is what makes the method quick. It turns a pair of simultaneous equations into two one-line calculations.

EXAMPLE 6.19

Express $\frac{x + 5}{x^2 - x - 2}$ in partial fractions.

This denominator is not factorised, so factorise it first:

$$x^2 - x - 2 = (x - 2)(x + 1).$$

Now proceed as before:

$$\frac{x + 5}{(x - 2)(x + 1)} = \frac{A}{x - 2} + \frac{B}{x + 1} \implies x + 5 = A(x + 1) + B(x - 2).$$

Setting $x = 2$: $7 = 3A$, so $A = \frac{7}{3}$. Setting $x = -1$: $4 = -3B$, so $B = -\frac{4}{3}$. Hence

$$\frac{x + 5}{x^2 - x - 2} = \frac{7}{3(x - 2)} - \frac{4}{3(x + 1)}.$$

TIP Examination questions usually give the denominator in expanded form. The factorising step is part of the answer, so write it out.

EXT Beyond the syllabus. Three things extend the method. None of them is on the AA HL syllabus, although the first is short enough to try in Your Turn Q30. A fraction whose numerator has degree equal to or higher than its denominator is **improper**, and must be divided first: $\frac{x^2}{x^2-1} = 1 + \frac{1}{x^2-1}$, and only the proper part is split. A **repeated** factor $(x-\alpha)^2$ needs one term for each power, $\frac{A}{x-\alpha} + \frac{B}{(x-\alpha)^2}$. An **irreducible quadratic**, one with negative discriminant, takes a linear numerator $\frac{Bx+C}{x^2+bx+c}$. One rule covers all of them: give each factor a numerator of degree one less than itself, and an extra term for every power of a repeated factor.

Factor in the denominator	Term(s) it contributes
linear $(x-\alpha)$	$\frac{A}{x-\alpha}$
repeated linear $(x-\alpha)^2$	$\frac{A}{x-\alpha} + \frac{B}{(x-\alpha)^2}$
irreducible quadratic (x^2+bx+c)	$\frac{Bx+C}{x^2+bx+c}$

YOUR TURN 6.6 Partial Fractions

27. Express each fraction in partial fractions.

(a) $\frac{1}{(x-1)(x-2)}$

(b) $\frac{5}{(x-2)(x+3)}$

(c) $\frac{3}{(x+1)(x+4)}$

(d) $\frac{x}{(x-1)(x+1)}$

28. Express each fraction in partial fractions.

(a) $\frac{2x+1}{x(x+1)}$

(b) $\frac{x+4}{x(x-2)}$

(c) $\frac{2x}{(x-3)(x+1)}$

(d) $\frac{5x-1}{(x-1)(x+2)}$

29. Factorise the denominator first, then express each fraction in partial fractions.

(a) $\frac{4x}{x^2-4}$

(b) $\frac{x+7}{x^2+x-2}$

30. Consider $\frac{x^2}{x^2-1}$.

(a) Explain why this fraction is improper.

(b) Divide to write it as $1 + \frac{1}{x^2-1}$.

(c) Hence express the whole fraction in partial fractions.

Reflections

TOK Abel and Galois proved something surprising. The roots of a general quintic *exist*, but they cannot be written using only addition, subtraction, multiplication, division and roots. The equation $x^5 - x - 1 = 0$ has a real solution. That number is completely determined, and a computer can calculate it to a thousand decimal places. Even so, no formula of that kind can express it exactly. What does it mean, then, to “know” a number? Is such a root any less real than one you can write down?

IM The history behind this chapter is unusual. In Italy in the 1530s, mathematicians competed in public problem-solving contests, and a loser could lose his university post. Methods were therefore kept secret. Niccolò Tartaglia found a way to solve the cubic and told no one. Gerolamo Cardano persuaded him to share it under an oath of silence, then published it in 1545, together with his student Ferrari’s solution of the quartic. Almost three hundred years passed before Niels Abel, who died of tuberculosis at twenty-six, proved that no such formula exists for the quintic. Évariste Galois, killed in a duel at twenty, explained exactly which equations can be solved by a formula and which cannot. To do this he created *group theory*, the mathematical study of symmetry, which is now used in particle physics and cryptography.

RW Polynomials are used constantly in computing, because they need only multiplication and addition. Those are the two operations a processor performs fastest. The curved shapes in fonts and animation are drawn with *Bézier curves*, which are polynomial curves whose shape is controlled by a few chosen points. *Error-correcting codes* are built from polynomials as well. These are methods of adding extra information to a message so that it can still be read after part of it is damaged, which is how a scratched disc still plays and how a faint signal from a spacecraft is recovered. Polynomials are also used to fit smooth models to scattered measurements in science and finance.

EXT The factor theorem raises a question it cannot answer. How many roots does a polynomial really have? Over the real numbers, $x^2 + 1$ has none. The **Fundamental Theorem of Algebra** gives the answer. Over the *complex* numbers of Chapter 17, every polynomial of degree n has exactly n roots, counted with multiplicity. The sum and product formulas then hold for every polynomial, because the roots they describe always exist.

End-of-Chapter Problems

1. Consider $f(x) = x^3 - 2x^2 - 5x + 6$.

- (a) Show that $x = 1$ is a root. [2]
- (b) Factorise $f(x)$ completely. [3]
- (c) Solve $f(x) = 0$. [1]

2. Consider $f(x) = 2x^3 + 3x^2 - 11x - 6$.

- (a) Show that $(x - 2)$ is a factor. [2]
- (b) Factorise $f(x)$ completely. [3]

3. Find the unknown coefficients in each polynomial.

- (a) When $x^3 + ax + b$ is divided by $(x - 1)$ the remainder is 4, and $(x + 1)$ is a factor. Find a and b . [5]
- (b) The polynomial $x^3 + ax^2 + bx - 2$ has $(x - 1)$ and $(x - 2)$ as factors. Find a and b . [5]
- (c) The remainder when $x^3 + cx + d$ is divided by $(x - 1)$ is 6, and $(x - 2)$ is a factor. Find c and d . [5]

4. The polynomial $x^3 - 3x^2 + kx - 8$ has $(x - 2)$ as a factor. Find k . [3]

5. Sketch each graph, showing all intercepts and describing the behaviour at each root.

- (a) $y = (x + 1)(x - 1)(x - 3)$ [4]
- (b) $y = x^2(x - 3)$ [4]

6. Build the equation described in each part.

- (a) A quadratic $x^2 + px + q$ has roots 2 and -5 . Find p and q . [3]
- (b) The cubic $x^3 + px^2 + qx + r$ has roots -1 , 2 and 4. Find p , q and r . [5]
- (c) The quadratic with roots α and β is $x^2 - 4x + 1$. Find a quadratic with leading coefficient 1 whose roots are $\alpha + 1$ and $\beta + 1$. [5]

7. Consider $f(x) = x^3 + 2x^2 - x - 2$.

- (a) Factorise $f(x)$ completely. [3]
- (b) Solve $f(x) = 0$. [1]

8. Solve each equation.

(a) $x^3 - 7x + 6 = 0$ [4]

(b) $2x^3 + 3x^2 - 3x - 2 = 0$ [5]

9. Each equation has roots α and β . Answer without solving the equation.

(a) $x^2 - 5x + 3 = 0$: find $\alpha^2 + \beta^2$. [4]

(b) $3x^2 - x - 2 = 0$: find $\frac{1}{\alpha} + \frac{1}{\beta}$. [3]

(c) $x^2 - 6x + 7 = 0$: find $\alpha^2 + \beta^2$ and $(\alpha - \beta)^2$. [5]

10. A cubic has roots 1, 2 and 3.

(a) Write it in factored and expanded form (leading coefficient 1). [3]

(b) State the sum and product of the roots and verify them from the expanded form. [2]

11. Consider $f(x) = x^4 - 5x^2 + 4$.

(a) Solve $f(x) = 0$. [3]

(b) State how many x -intercepts the graph has. [1]

12. Show that $x = 1$ is a root of $x^3 + 4x^2 + x - 6 = 0$, and find the other roots. [4]

13. The equation $x^2 - kx + 9 = 0$ has equal roots. Use the sum and product of roots to find k . [4]

14. Consider $f(x) = 2x^3 - x^2 - 13x - 6$.

(a) Show that $(x + 2)$ is a factor. [2]

(b) Factorise $f(x)$ completely and solve $f(x) = 0$. [4]

15. The polynomial $x^3 - kx + 2$ has $(x - 2)$ as a factor.

(a) Find k . [2]

(b) Find the other two roots, in exact form. [4]

16. The polynomial $x^3 - 2x^2 + x - 2$ factorises as $(x - 2)(x^2 + 1)$.

(a) Verify this by expansion. [2]

(b) State the only real root, and explain why there are no others. [2]

17. When a polynomial $p(x)$ is divided by $(x - 1)$ the remainder is 3, and by $(x - 2)$ the remainder is 5. Find the remainder when $p(x)$ is divided by $(x - 1)(x - 2)$. [6]

18. The roots of $x^3 - 6x^2 + 11x - 6 = 0$ are α, β, γ . Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the

equation. [5]

19. Prove that if a polynomial with integer coefficients has a rational root $\frac{p}{q}$ in lowest terms, then p divides the constant term and q divides the leading coefficient. [6]

20. The cubic $x^3 + px + q = 0$ has a repeated root. Show that $4p^3 + 27q^2 = 0$. [6]

21. Find all real values of k for which $x^3 - 3x + k = 0$ has three distinct real roots. (*Hint: use Q20 to find the values of k that give a repeated root*) [6]

22. Consider $f(x) = 2x^3 - 3x^2 - 11x + 6$.

- (a) Show that $(x - 3)$ is a factor of $f(x)$. [2]
- (b) Factorise $f(x)$ completely. [4]
- (c) Solve $f(x) = 0$. [2]
- (d) State the y -intercept, and describe the behaviour of $f(x)$ as $x \rightarrow \pm\infty$. [3]
- (e) Sketch $y = f(x)$, showing all intercepts. [2]
- (f) Hence solve the inequality $f(x) \leq 0$. [3]

23. The polynomial $p(x) = x^3 + ax^2 + bx + 6$ leaves a remainder of 12 when divided by $(x - 2)$, and $(x + 1)$ is a factor.

- (a) Show that $4a + 2b = -2$ and $a - b = -5$. [4]
- (b) Hence find a and b . [2]
- (c) Factorise $p(x)$ as far as possible over the real numbers. [4]
- (d) State how many real roots $p(x) = 0$ has, giving a reason. [2]
- (e) Find the remainder when $p(x)$ is divided by $(x - 1)(x + 1)$. (*Hint: the remainder has the form $rx + s$*) [4]

24. The cubic equation $x^3 + px^2 + qx + r = 0$ has roots α , β and γ .

- (a) Write down $\alpha + \beta + \gamma$, $\alpha\beta + \beta\gamma + \gamma\alpha$ and $\alpha\beta\gamma$ in terms of p , q and r . [3]
- (b) The equation $x^3 - 6x^2 + 3x + 10 = 0$ has roots α , β , γ . Find $\alpha^2 + \beta^2 + \gamma^2$ without solving the equation. [4]
- (c) Find $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. [3]
- (d) Given that one root is -1 , find the other two. [3]
- (e) Write down a cubic equation with integer coefficients whose roots are 2α , 2β and 2γ . [3]

25. Express each fraction in partial fractions.

- (a) $\frac{7x - 1}{(x - 1)(x + 2)}$ [4]
- (b) $\frac{1}{x^2 - 4}$ [4]

(c) $\frac{x+5}{x^2+x}$ [4]

Challenge Questions

26. Power sums of the roots. The equation $x^2 - px + q = 0$ has roots α and β . For each integer $k \geq 0$ define

$$S_k = \alpha^k + \beta^k.$$

- (a) Write down the values of S_0 and S_1 . [2]
 (b) Show that $S_2 = p^2 - 2q$. [2]
 (c) Find S_3 in terms of p and q . [3]
 (d) Explain why $\alpha^k = p\alpha^{k-1} - q\alpha^{k-2}$ for every $k \geq 2$, and deduce that $S_k = pS_{k-1} - qS_{k-2}$. [4]
 (e) The equation $x^2 - 3x + 1 = 0$ has roots α and β . Use the recurrence to find S_2, S_3 and S_4 . [4]
 (f) The roots of $x^2 - 3x + 1 = 0$ are irrational. Explain why S_k is nevertheless an integer for every k . [3]

27. Palindromic polynomials. A polynomial is *palindromic* if its list of coefficients reads the same forwards as backwards. For example $x^4 + x^3 - 4x^2 + x + 1$ has coefficients 1, 1, -4, 1, 1.

- (a) Explain why $x = 0$ is never a root of a palindromic polynomial. [2]
 (b) Let $P(x)$ be palindromic of degree n . Show that $x^n P\left(\frac{1}{x}\right) = P(x)$, and deduce that if α is a root of P then so is $\frac{1}{\alpha}$. [4]
 (c) Divide $x^4 + x^3 - 4x^2 + x + 1 = 0$ through by x^2 and substitute $u = x + \frac{1}{x}$. Show that the equation becomes $u^2 + u - 6 = 0$. [4]
 (d) Hence find all four roots of $x^4 + x^3 - 4x^2 + x + 1 = 0$ in exact form. [5]
 (e) Show that every palindromic polynomial of odd degree has $x = -1$ as a root. [3]

28. Splitting a sum with partial fractions. Partial fractions are usually a step towards integration. This investigation uses them for something else: adding infinitely many terms exactly.

- (a) Show that $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ for every positive integer n . [2]
 (b) Hence write out the sum $\sum_{n=1}^N \frac{1}{n(n+1)}$ term by term, and show that almost every term cancels. Deduce that the sum equals $\frac{N}{N+1}$. (Hint: write the first three and the last two

terms)

[4]

- (c) State the value the sum approaches as $N \rightarrow \infty$, and explain what that value means for the graph of the partial sums. [2]
- (d) Show that $\frac{1}{n(n+2)} = \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+2} \right)$, and hence find $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$. [4]
- (e) Investigate $\sum_{n=1}^{\infty} \frac{1}{n(n+k)}$ for a positive integer k . Find its value for $k = 3$ and $k = 4$, then state a general formula and justify it. [6]

